MO-201 Materials Science Tutorial 2 (Odd Sem 2026-27)

Q.1 Calculate the density of MgO from the following data.
Structure: FCC anion packing, cations in the octahedral voids.
Radius of Mg^{2+} ion = 0.78 Å
Radius of O^{2-} ion = 1.32 Å

Solution

Latice Parameter a,

$$a = 2(r_{Mg^{2+}} + r_{O^{2-}}) = 2(0.78 + 1.32) = 4.20\text{Å}$$

The effective number of oxygen anions at FCC positions in the unit cell $= 8 \times 1/8$ (corner ions) $+6 \times 1/2$ (face centred ions) $= 4$.

The effective number of magnesium cations in the octahedral voids $= 1$ (body centre) $+12 \times 1/4$ (midpoints of cube edges) $= 4$.

$$\text{Density} = \frac{\text{Mass of atoms in unit cell}}{\text{Volume of unit cell}} = \frac{(16 + 24.3) \times 1.660 \times 10^{-27} \times 4}{4.20^3 \times 10^{30}} = 3610 \text{ kg m}^{-3} = 3.61 \text{ g cm}^{-3}$$

Q.2 The following diffractometer data (expressed as 2θ) were generated from a specimen irradiated with silver (Cu) K radiation: 38.40°; 44.50°; 64.85°; 77.90°; 81.85°; 98.40°; 111.20°. λ= 1.5418 Å

(a) Determine the crystal structure.

(b) Calculate the lattice constant, a.

(c) Assume that the crystal is a pure metal and on the basis of the hard-sphere approximation calculate the atomic radius.

Solution

Follow the procedure suggested in lecture:

(1) Start with 2θ values and generate a set of sin2 θ values.

(2) Normalize the sin2 θ values by generating $\sin^2\theta_n/\sin^2\theta_1$.

(3) Clear fractions from the "normalized" column.

(4) Speculate on the hkl values that would seem as $h^2 + k^2 + l^2$ to generate the sequence of the "clear fractions" column.

(5) Compute for each θ the value of $\frac{\sin^2 \boldsymbol{\theta}}{h^2+k^2+l^2}$ on the basis of the assumed hkl values. If each entry in this column is identical, then the entire process is validated.

(a).
For the data set in question, it is evident from the hkl column that the crystal structure is FCC (see table below).

(b)

$$\frac{\lambda^2}{4a^2} = \frac{\sin^2\theta}{h^2 + k^2 + l^2} = 0.0358$$

$$\lambda_{Cu K_\alpha} = 1.5418\text{Å}, \quad \therefore \ a = \frac{1.5418}{(4 \times 0.0358)^{1/2}} = 4.07\text{Å}$$

(c) Since this is FCC, $a\sqrt{2} = 4r$

$$r = \frac{a\sqrt{2}}{4} = 4.07 \times \frac{\sqrt{2}}{4} = 1.44 \text{ Å}$$

2θ	$\sin^2 \theta$	$\sin^2\theta_n/\sin^2\theta_1$	clear fractions	(hkl)	$\frac{\sin^2 \theta}{h^2 + k^2 + l^2}$
38.40	0.108	1.00	3	111	0.0360
44.50	0.143	1.32	4	200	0.0358
64.85	0.288	2.67	8	220	0.0359
77.90	0.395	3.66	11	311	0.0358
81.85	0.429	3.97	12	222	0.0358
98.40	0.573	5.31	16	400	0.0358
111.20	0.681	6.31	19	331	0.0358

Q.3 The first order diffraction by a crystal plane having d= 2.34 Å in a solid is observed at the Bragg angle of 2θ=30°.

Using the same radiation and first order diffraction,2θ =60° for another solid. Calculate the d value for the 2nd solid.

Solution

$n\lambda = 2d\sin\theta$
For first Solid
Given data, $n = 1$

$d = 2.34\ A^{\circ}$
$\theta = 15^{\circ}$

Therefore, $\lambda = 1.211$ Å

For second solid,

Given data,
$$n = 1$$
$$\lambda = 1.211\ A^{\circ}$$
$$\theta = 30^{\circ}$$
$$1 \times 1.211\ A^{\circ} = 2 \times d \times \sin\ 30^{\circ}$$
$$1.211\ A^{\circ} = 2 \times d \times 0.5$$
$$d = 1.211\ A^{\circ}$$

Q.4 The spacing between the principal planes in a crystal of NaCl is 2.82 Å. It is found that the first order Bragg's reflection occurs at 20°.

a) What is the wavelength of X-rays?

b) At what angle, the second order reflection occurs?

c) What is the highest order of reflection seen?

Solution

$\mathrm{d} = 2.82\ \mathrm{A}^0 = 2.82 \times 10^{-10}\ \mathrm{m},\ \mathrm{n} = 1,\ 2\theta = 20°, \theta =\ 10°$

a) According to Bragg's Law of X-ray Diffraction $2d\sin\ \theta = n\lambda$

$$\therefore \lambda = \frac{2\ \mathrm{d}\sin\ \theta}{\mathrm{n}} = \frac{2 \times 2.82 \times 10^{-10} \times \sin\ 10}{1} = 9.79 \times 10^{-11}\ \mathrm{m} = 0.979\ \mathrm{A}^0$$

b) For $n = 2\ \therefore \theta = \sin^{-1}\left(\frac{\mathrm{n}\lambda}{2\ \mathrm{d}}\right) = \sin^{-1}\left(\frac{2\times9.79\times10^{-11}}{2\times2.82\times10^{-10}}\right) = 20.31^0$

c) For highest order, maximum value of $\sin\ \theta = 1$

$$\therefore \mathrm{n_{max}} = \frac{2\ \mathrm{d}(\sin\ \theta)_{\mathrm{max}}}{\lambda} = \frac{2 \times 2.82 \times 10^{-10} \times 1}{9.79 \times 10^{-11}} = 5.76$$

As n is an integer, highest order than can be seen is 5